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Agenda:

1. What is Data Science?
2. How Visualization plays a critical role in DS world?
3. Tools in Data Visualization - Tableau/ Power BI
4. Intro to Tableau / Versions of tableau
5. Installing tableau
6. Loading the data
7. Case Study to help Australian Wine Company CEO with timely Insights
8. Visualization vs Story Telling
9. Dimensions and measures in Tableau
10. Building Charts – Demo
11. Building Advance charts for forecasting
12. Building an interactive dashboard
13. Deploying the Dashboard
14. Q&A and Posting it on Linkedin

1. What is Data Science?

Data Science is the interdisciplinary field that converts raw data into actionable business insights using:

- Statistics and probability
- Data analysis and visualization
- Business domain understanding

- Technology and tools

Data Science Lifecycle

- Business Problem Definition
- Data Collection
- Data Cleaning & Preparation
- Exploratory Data Analysis (EDA)
- Visualization & Insight Generation
- Decision-Making & Communication

👉 Visualization is the bridge between analysis and decision-making.

2. Role of Visualization in the Data Science World

- Visualization answers business questions faster than tables or raw numbers.
- Why Visualization is Critical
- Humans process visuals faster than text
- Helps identify trends, patterns, and outliers
- Enables stakeholder-friendly communication
- Drives data-driven decisions, not intuition

Example

Table: Monthly sales numbers
Chart: Immediate visibility of seasonality and decline

📌 "Without visualization, data science remains incomplete."

3. Tools Used in Data Visualization

Tool	Usage
Excel	Basic charts, quick analysis
Python / R	Programmatic visualization
Power BI	Microsoft ecosystem

Tableau	Best for interactive dashboards & storytelling

Why Tableau?

- Drag-and-drop interface
- No coding required
- Strong interactivity
- Enterprise-level dashboards
- Excellent storytelling capabilities

4. Introduction to Tableau & Versions

What is Tableau?

Tableau Desktop is a BI tool used to analyze data visually and build interactive dashboards.

Version	Purpose
Tableau Desktop	Development & analysis
Tableau Public	**Free, public sharing**
Tableau Server	Enterprise hosting
Tableau Cloud	Cloud-based deployment

Tableau Reader	View-only

5. Installing Tableau

Installation Steps

1. Download Tableau Public - https://www.tableau.com/products/public/download
2. Install and Launch Tableau Public
3. Choose data source

📌 Tableau supports Excel, CSV, SQL, cloud sources.

6. Loading the Data (Global Superstore)

Dataset Used - Global Superstore

Global Superstore Dataset

Contains:

- Orders
- Customers
- Products
- Regions
- Sales, Profit, Quantity, Discount

Steps

1. Open Tableau Desktop
2. Connect → Microsoft Excel
3. Select *Global Superstore.xlsx*
4. Drag Orders sheet to canvas
5. Move to Worksheet

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7. Case Study: Global Superstore CEO Dashboard

🎯 Business Context

Global Superstore is a multinational retail company selling office supplies, furniture, and technology products across multiple regions and markets.

The **CEO** is facing the following strategic challenges:

- Revenue is growing, but **profitability is inconsistent**
- Some regions show **high sales but poor margins**
- Discounting strategies are unclear and may be eroding profits
- Leadership needs **forward-looking insights**, not just historical reports
- Decisions must be made **quickly**, without waiting for static reports

The CEO has requested a **single interactive dashboard in Tableau** that provides:

- A **high-level executive overview**
- The ability to **drill down** into problem areas
- **Forecasted insights** to support planning and budgeting

🎯 Primary Business Objectives

The dashboard should enable the CEO to:

1. **Monitor overall business health**

    - Track sales, profit, and growth trends over time
2. **Identify problem areas**
    - Spot loss-making regions, markets, and products
3. **Optimize product and pricing strategy**
    - Understand which categories and sub-categories drive or destroy profit
4. **Evaluate discount effectiveness**
    - Assess whether discounts are increasing volume at the cost of profit
5. **Support strategic planning**
    - Use sales forecasts to guide inventory and marketing decisions

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🔍 Business Questions

1️⃣ How are sales and profits trending over time?

Business Need

- The CEO wants to understand whether the company is:
    - Growing sustainably
    - Experiencing seasonal fluctuations
    - Improving or declining in profitability

Analysis Approach in Tableau

- Line charts showing:
    - Sales over time
    - Profit over time
- Year-over-year and month-over-month trends

Insight Expectation

- Identify periods of strong growth or decline
- Detect gaps where sales increased but profits did not
- Understand seasonality patterns (e.g., end-of-year spikes)

2️⃣ Which regions and markets are most profitable?

Business Need

- Expansion and investment decisions depend on **profitability**, not revenue alone.
- The CEO wants to know:
  - Where to invest more
  - Where to reduce costs or rethink strategy

Analysis Approach in Tableau

- Bar charts or maps showing:
  - Sales by Region
  - Sales by Market
- Color-coded performance (profit vs loss)

Insight Expectation

- Identify high-performing regions
- Detect regions with negative profit despite strong sales
- Support decisions on market expansion or exit

3️⃣Which products and sub-categories are causing losses?

Business Need

- Not all products contribute equally to business success.
- Some products may:
  - Have high return rates
  - Require heavy discounting
  - Carry high logistics costs

Analysis Approach in Tableau

- Tree maps or bar charts showing:
  - Profit by Category and Sub-Category
- Drill-down capability from category → sub-category → product

Insight Expectation

- Identify loss-making products
- Spot sub-categories that consistently reduce margins
- Support decisions on product discontinuation or repricing

4️⃣ Are discounts impacting profitability?

Business Need

- Discounts are used to drive sales, but may be:
    - Reducing margins excessively
    - Encouraging low-value customers

Analysis Approach in Tableau

- Scatter plots:
    - Discount vs Profit
    - Discount vs Sales
- Trend lines to identify correlation

Insight Expectation

- Understand if higher discounts correlate with lower profit
- Identify optimal discount ranges
- Support pricing and promotion strategy adjustments

5️⃣ What will sales look like in the next 6 months?

Business Need

- The CEO needs **forward-looking insights** for:
    - Inventory planning
    - Staffing
    - Marketing spend
    - Budget forecasting

Analysis Approach in Tableau

- Time-series forecasting using historical sales data
- Confidence intervals to show uncertainty

Insight Expectation

- Predict future demand
- Identify expected high and low sales periods
- Enable proactive decision-making

🎯 Final Outcome for the CEO

By using this Tableau dashboard, the CEO will be able to:

- Move from **reactive reporting** to **proactive decision-making**
- Quickly identify risks and opportunities
- Ask better strategic questions in leadership meetings
- Align sales, marketing, and operations around data-backed insights

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8. Dimensions and Measures in Tableau

In **Tableau**, every field in your dataset is classified as either a **Dimension** or a **Measure**. This classification determines **how Tableau treats the data** and **how it appears visually**.

Tableau Color Coding

- **Blue pills → Dimensions**
- **Green pills → Measures**

This color difference is fundamental and appears consistently across:

- Data Pane
- Rows and Columns shelves
- Marks card

🔵 Dimensions (Categorical Fields)

What are Dimensions?

Dimensions are fields that **describe or categorize data**.
They answer questions like **who, what, where, and which**.

📌 Tableau treats dimensions as **discrete values** and displays them in **blue**.

Key Characteristics

- Typically text or date fields
- Create headers, labels, and segments
- Used to **slice, group, or filter data**
- Do NOT perform mathematical aggregation by default

Global Superstore Examples

Dimension	Business Meaning
Region	Geographic grouping (East, West, APAC)
Country	Country-level analysis
Category	High-level product grouping
Sub-Category	Detailed product classification
Customer Segment	Consumer, Corporate, Home Office

How Dimensions Work in Tableau

When you drag a **dimension** to:

- **Rows / Columns** → Tableau creates separate headers
- **Filters** → Data is sliced
- **Color / Label** → Data is segmented

Example

- Drag **Region (blue)** to Rows
- Drag **Sales (green)** to Columns

👉 Tableau shows **sales broken down by each region.**

🟢 Measures (Numerical Fields)

What are Measures?

Measures are fields that **quantify performance**.
They answer **how much, how many, and how often**.

📌 Tableau treats measures as **continuous values** and displays them in **green**.

Key Characteristics

- Always numeric
- Aggregated by default (SUM, AVG, COUNT)
- Used to evaluate business performance
- Drive chart axes and trends

Global Superstore Examples

Measure	Business Meaning
Sales	Revenue generated
Profit	Business profitability
Quantity	Units sold
Discount	Price reduction offered

How Measures Work in Tableau

When you drag a **measure** to:

- **Rows / Columns** → Creates axes
- **Color / Size** → Shows magnitude
- **Label** → Displays numeric values

Example

- Drag **Sales (green)** to Columns

👉 Tableau automatically uses **SUM(Sales)**.

🔄 Blue vs Green in Tableau (Key Difference)

Feature	Dimension (Blue)	Measure (Green)

Data Type	Categorical	Numerical
Color	Blue	Green
Purpose	Slice & group	Measure performance
Aggregation	No	Yes (SUM, AVG, etc.)
Example	Region, Category	Sales, Profit

📌 The Golden Rule in Tableau

> **Dimensions slice the data.**
> **Measures measure performance.**

Practical Example

- **Dimension:** Region → "Where?"
- **Measure:** Profit → "How well?"

Can a field change color?

Yes.

- A **date field** can be:
  - Blue (Discrete) → Month names
  - Green (Continuous) → Timeline

Can a number be a dimension?

Yes.

- Example: **Customer ID**
  - Numeric, but categorical
  - Used for grouping, not calculation

🎯 Business Interpretation Example

Question:
Which **Sub-Category** is causing losses?

- **Sub-Category (Blue Dimension)** → breaks data
- **Profit (Green Measure)** → evaluates impact

👉 Tableau visually answers the business question.

Summary

- **Blue = Dimensions → break data into groups**
- **Green = Measures → evaluate performance**

Below is a **clear, business-aligned chart recommendation** for **each CEO question in the Global Superstore case study**, explicitly mapping **business intent → Tableau chart type → dimensions & measures → expected insight**. This is ideal for teaching, demos, and dashboard design.

Chart Recommendations Based on Global Superstore Case Study

🟨 Data Storytelling session

1️⃣How are sales and profits trending over time?

🎯 Business Intent

Understand growth, seasonality, and whether profit is keeping pace with sales.

📊 Recommended Chart

Dual Line Chart (Time-Series)

Tableau Configuration

- **Columns:** Order Date (Continuous – Year / Month)
- **Rows:**
  - SUM(Sales)
  - SUM(Profit) (Dual Axis)
- **Marks:** Line
- **Color:** Sales vs Profit

Why This Chart?

- Clearly shows long-term trends
- Highlights divergence between sales and profit
- Helps identify seasonal patterns

Insight Example

Sales may rise steadily while profit fluctuates, indicating pricing or cost issues.

2️⃣ Which regions and markets are most profitable?

🎯 Business Intent

Identify where the company should invest more or intervene.

📊 Recommended Chart

Horizontal Bar Chart or **Filled Map**

Tableau Configuration (Bar Chart)

- **Rows:** Region / Market
- **Columns:** SUM(Profit)
- **Color:** Profit (Red–Green Diverging)
- **Sort:** Descending by Profit

Why This Chart?

- Easy comparison across regions
- Instantly shows profit vs loss areas

Insight Example

A region with high sales but low profit becomes immediately visible.

3️⃣Which products or sub-categories are causing losses?

🎯 Business Intent

Optimize product portfolio and eliminate margin leakage.

📊 Recommended Chart

Tree Map (Category → Sub-Category)

Tableau Configuration

- **Size:** SUM(Sales)
- **Color:** SUM(Profit)
- **Label:** Sub-Category

Why This Chart?

- Combines volume and profitability
- Quickly highlights loss-making products
- Excellent for executive audiences

Insight Example

Large red blocks indicate high-revenue but loss-making sub-categories.

4️⃣Are discounts impacting profitability?

🎯 Business Intent

Evaluate whether discounting strategy is effective or harmful.

📊 Recommended Chart

Scatter Plot with Trend Line

Tableau Configuration

- **Columns:** Discount
- **Rows:** Profit
- **Detail:** Sub-Category or Product Name
- **Analytics:** Trend Line

Why This Chart?

- Shows correlation between discount and profit
- Identifies optimal discount ranges

Insight Example

Higher discounts may correlate with negative profit, indicating over-discounting.

5️⃣What will sales look like in the next 6 months?

🎯 Business Intent

Support planning, budgeting, and inventory decisions.

📊 Recommended Chart

Forecasted Line Chart

Tableau Configuration

- **Columns:** Order Date (Continuous)
- **Rows:** SUM(Sales)
- **Analytics:** Forecast (6 months)
- **Confidence Interval:** Enabled

Why This Chart?

- Provides forward-looking insights
- Helps anticipate demand fluctuations

Insight Example

Seasonal spikes can be anticipated and planned for in advance.

🧩 Executive Dashboard Layout

Section	Chart
KPI Overview	Total Sales, Total Profit, Profit Margin
Trends	Dual Line Chart (Sales & Profit)
Geography	Profit by Region
Products	Tree Map
Pricing Impact	Scatter Plot
Future Outlook	Forecast Chart

12. Building an Interactive Dashboard

Dashboard Components

- KPIs: Sales, Profit, Orders
- Trend charts
- Region filters
- Category selectors

Interactivity Features

- Filters
- Highlight actions
- Tooltips
- Dashboard actions

📌 Goal: CEO should **explore insights without asking analysts**

13. Deploying the Dashboard

Deployment Options

- Tableau Public (for learning & LinkedIn)
- Tableau Server / Cloud (enterprise)

Steps (Tableau Public)

1. Save workbook
2. Publish to Tableau Public
3. Get shareable link

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14. Key Insights from the Case Study

- Technology category drives sales but has profit volatility
- Some regions generate losses despite high revenue
- Excessive discounts negatively impact profit

- Forecast shows seasonal spikes

📌 **Actionable Recommendation**:

- Reduce discounts in loss-making regions
- Focus on profitable sub-categories
- Align inventory with forecasted demand

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15. Q&A and Posting on LinkedIn

How to Post on LinkedIn

- Screenshot dashboard
- Share Tableau Public link

Post:

📊 Turning Data into Decisions with Tableau

Excited to share an **interactive executive dashboard built using Tableau** as part of a **Global Superstore case study**.

🔍 **Business Problem Addressed**
The CEO of a global retail organization needed **timely, data-driven insights** to:

- Improve profitability
- Identify loss-making regions and products
- Evaluate the impact of discounts
- Forecast future sales for better planning

📈 **What the Dashboard Delivers**

- Sales and profit trends over time
- Region- and market-wise Sales analysis
- Identification of loss-making sub-categories

- Discount vs profit analysis
- 6-month sales forecast to support strategic decisions

🛠️ **Tool Used:** Tableau
🌍 **Dataset:** Global Superstore

This exercise reinforces an important lesson:

> Data visualization is not about charts — it is about **business storytelling and decision-making**.

Please post your dashboard screenshot and Tableau Public link, and **tag**:

- **Saurabh Kango** ⟶ https://www.linkedin.com/in/saurabhkango/
- **Scaler** ⟶ https://www.linkedin.com/school/scalerofficial/posts/?feedView=all

Let's showcase how analytics skills translate into real-world business impact.

Add hashtags:
`#DataVisualization #Tableau #Analytics #BusinessIntelligence #LearningByDoing #DashboardDesign #DataStorytelling`

📌 This builds **professional credibility**.